Comparative analysis of the *Eulemur* vocal repertoire: a dynamic time warping approach.

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Appendix S4 *Dendrogram Topology*

Dendrogram Description

Eulemur fulvus

A Territorial Advertisement Call variant segregated first (Fig. 1). Another first-level clade led to two second-level clades. One of these second-level clades showed two third-level leaves (Grunt-Tonal Call and a Scream variant) and two fourth-level clades (two Grunt variants; a Scream variant and Long Grunt-Tonal Call). Another second-level clade led to two bifolious third-level clades (Chatter and Group Cohesion Call; two Territorial Advertisement Call variants).

Eulemur rufus

The analysis showed two first-level clades (Fig. 2). In one of them, we found a leaf corresponding to a Territorial Advertisement Call variant and a second-level clade including two more Territorial Advertisement Call variants. The other first-level clade branches into two second-level clades. One featured Grunt, Grunted Hoot, and a Scream variant; the other indicated four (third-level) clades where two Scream variants are more similar to each other than they are to Alarm Call and Group Cohesion Call.

Eulemur rubriventer

The cluster dendrogram showed two first-level branches (Fig. 3), both originating two second-level branches. The second-level branches led to leaves indicating that two Hoot-Tonal Call variants are more similar to each other than they are to Tonal Call and Long Grunt. The other first-level clade ended in groups of leaves indicating on one side three Snort-Grunt-Tonal Call variants and two Tonal Call variants, and on the other side two Scream variants, Gurgle, Grunt, and Hoot. According to the tree structure, we found that in the (third-level) clade including Gurgle, Grunt, and Hoot, the latter two are more similar to each other than they are to Gurgle.

Eulemur mongoz

The cluster analysis identified two first-level clades (Fig. 4). One led to a second-level leaf indicating Group Cohesion Call (usually referred to as Croui-Croui) and a third-level clade showing leaves for Alarm Long Grunt and Long Grunt. Two second-level clades appeared in the other first-level branch. Each of them branched into a leaf (Territorial Advertisement Call; Grunted Hoot) and a third-level clade (Scream and Grunt-Tonal Call; Click and Grunt).

Eulemur coronatus

The clustering analysis (Fig. 5) indicated two first-level clades. A first-level clade branches into two bifolious clades, one showing two Alarm Call variants and the other identifying an Alarm Call and a Scream variant. The other first level-clade branches into two clades, then into three third-level clades (ending in Grunt-Tonal Calls, three Grunt variants, and two Scream variants, and Tonal Call, Alarm Call) and a leaf (a Scream variant).

Eulemur flavifrons

The analysis identified two first-level branches (Fig. 6). One led to a second-level branch and two third-level branches, indicating four leaves (a Long Grunt-Tonal Call variant, Grunt-Tonal Call, and two Grunted Hoots variants). A second-level leaf ended with Long Grunt-Tonal Call. Another first-level branch split into two bifolious clades, where all leaves indicated Alarm Call variants.

Eulemur macaco

A first-level clade ended with two leaves corresponding to Alarm Call and a Scream variant (Fig. 7). Another first-level clade split into two second-level clades, each of them originating two third-level clades. We found that in one of these third-level clades, Grunt is more similar to Tonal Call than to two other variants of Scream. The other two further variants of Scream were more similar to Long Grunt-Tonal Call and Long Grunt than to each other.

Appendix S5 ***We reported the distribution of the vocalizations according to the*** **a priori** ***classification and the cluster membership emerging from the cluster analysis.***

Eulemur fulvus

Table SII. Distribution of the vocalizations of *Eulemur fulvus* in the clusters identified using the affinity propagation algorithm.

Cluster ID	AC	CH	CL	CO	H	GH	GR	GT	LGTC	SC	SN	TE	TO	TG	TOT
1	1	1								8					**10**
2		1	5		2	3	13				7		1		**32**
3		1		1								6			**8**
4					3	11	37	7			4		5	14	**81**
5	1			3											**4**
6												4			**4**
7									11						**11**
8		5					1	8							**14**
9	3	6		1					2	1					**13**
10												4			**4**
11										9					**9**
Input (N)	**5**	**14**	**5**	**5**	**5**	**14**	**51**	**15**	**13**	**18**	**11**	**14**	**6**	**14**	**190**
Exemplars (N)	**0**	**1**	**0**	**1**	**0**	**0**	**2**	**1**	**1**	**2**	**0**	**3**	**0**	**0**	**11**

Columns indicate the result of the *a priori* classification (AC = Alarm Call, CH = Chatter, CL = Click, CO = Group Cohesion Call, H = Hoot, GH = Grunted Hoot, GR = Grunt, GT = Grunt-Tonal Call, LGTC = Long Grunt-Tonal Call, SC = Scream, SN = Snort, TE = Territorial Advertisement Call, TO = Tonal Call, TG = Tonal Call-Grunt), while rows show cluster membership and composition. The number of vocalizations in a cluster (TOT), the number of vocalizations corresponding to an *a priori* vocal type (Input), and the number of exemplars chosen from that vocal type during the cluster analysis (Exemplars) are also reported.

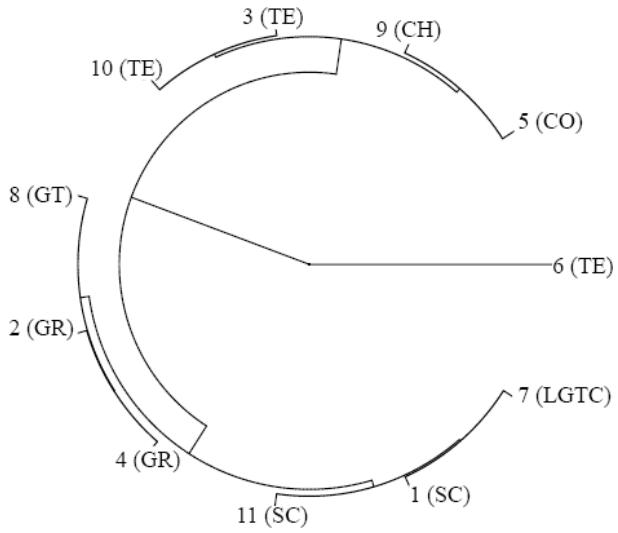


Fig. 9 Schematic representation of the polar dendrogram in Fig. 1 (*E. fulvus*), showing Cluster ID and abbreviation of the vocal type corresponding to the chosen exemplar

Eulemur rufus

Table SIII. Distribution of the vocalizations of *Eulemur rufus* in the clusters identified using the affinity propagation algorithm.

Cluster ID	AC	CL	CO	H	GH	GR	GT	LGA	SC	SN	TE	TO	TOT
1	8		1					2					**11**
2	1		5						1				**7**
3		12		11	2	24				12		1	**62**
4			2								11		**13**
5								2	5				**7**
6								2	7				**9**
7			3								4		**7**
8					27	8	11					13	**59**
9											8		**8**
10								2	5				**7**
Input (N)	**9**	**12**	**11**	**11**	**29**	**32**	**11**	**8**	**18**	**12**	**23**	**14**	**190**
Exemplars (N)	**1**	**0**	**1**	**0**	**1**	**1**	**0**	**0**	**3**	**0**	**3**	**0**	**10**

Columns indicate the result of the *a priori* classification (AC = Alarm Call, CL = Click, CO = Group Cohesion Call, H = Hoot, GH = Grunted Hoot, GR = Grunt, GT = Grunt-Tonal Call, LGA = Alarm Long Grunt, SC = Scream, SN = Snort, TE = Territorial Advertisement Call, TO = Tonal Call), while rows show cluster membership and composition. The number of vocalizations in a cluster (TOT), the number of vocalizations corresponding to an *a priori* vocal type (Input), and the number of exemplars chosen from that vocal type during the cluster analysis (Exemplars) are also reported.

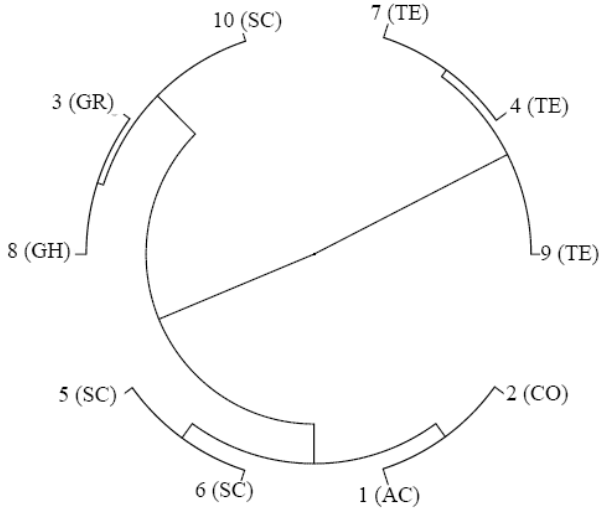


Fig. 10 Schematic representation of the polar dendrogram in Fig. 2 (*E. rufus*), showing Cluster ID and abbreviation of the vocal type corresponding to the chosen exemplar

Eulemur rubriventer

Table SIV. Distribution of the vocalizations of *Eulemur rubriventer* in the clusters identified using the affinity propagation algorithm.

Cluster ID	CH	CL	H	HT	GH	GR	GT	GU	LG	SC	SG	SGTC	SN	TO	TOT
1										12					**12**
2				1										4	**5**
3							2		1			4			**7**
4				5		1	2								**8**
5			2			8					3	2	3		**18**
6		6	10		5	7	1			3	5		5		**42**
7							2	3	1					7	**13**
8	2			3											**5**
9										6				2	**8**
10	1						4		2		1	5			**13**
11	3			2			1		4			4			**14**
12				2		2	3							8	**17**
13									3						**3**
14						8	5	8						6	**27**
Input (N)	**6**	**6**	**12**	**13**	**5**	**26**	**20**	**11**	**11**	**21**	**9**	**15**	**8**	**27**	**190**
Exemplars(N)	**0**	**0**	**1**	**2**	**0**	**1**	**0**	**1**	**1**	**2**	**0**	**3**	**0**	**3**	**14**

Columns indicate the result of the *a priori* classification (CH = Chatter, CL = Click, H = Hoot, HT = Hoot-Tonal Call, GH = Grunted Hoot, GR = Grunt, GT = Grunt-Tonal Call, GU = Gurgle, LG = Long Grunt, SC = Scream, SG = Snort-Grunt, SGTC = Snort-Grunt-Tonal Call, SN = Snort, TO = Tonal Call), while rows show cluster membership and composition. The number of vocalizations in a cluster (TOT), the number of vocalizations corresponding to an *a priori* vocal type (Input), and the number of exemplars chosen from that vocal type during the cluster analysis (Exemplars) are also reported.

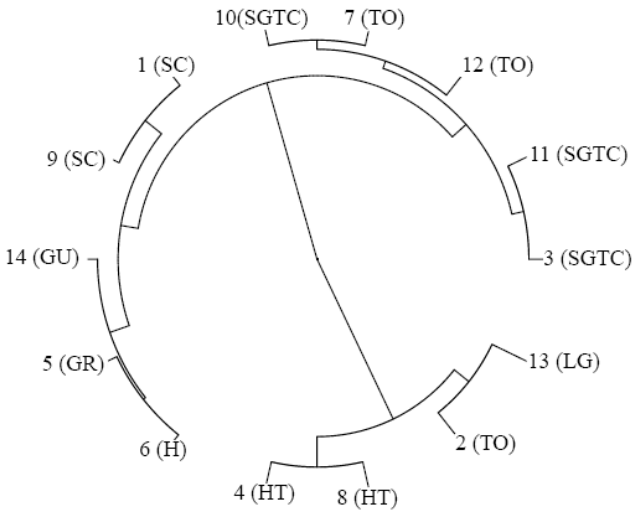


Fig. 11 Schematic representation of the polar dendrogram in Fig. 3 (*E. rubriventer*), showing Cluster ID and abbreviation of the vocal type corresponding to the chosen exemplar

Eulemur mongoz

Table SV. Distribution of the vocalizations of *Eulemur mongoz* in the clusters identified using the affinity propagation algorithm.

Cluster ID	AC	CH	CL	CO	H	GH	GR	GT	LG	LGA	SC	SN	TE	TO	TOT
1				11					1						**12**
2		5	2		5	5	4				2			1	**24**
3								1	1	12					**14**
4		3								3	5				**11**
5									5	1					**6**
6		2				1	17	1							**21**
7						1		6		2			1		**10**
8		1						5	2	5			10		**23**
9	9		16		15		8					13		8	**69**
Input (N)	**9**	**11**	**18**	**11**	**20**	**7**	**29**	**13**	**9**	**23**	**7**	**13**	**11**	**9**	**190**
Exemplars (N)	**0**	**0**	**1**	**1**	**0**	**1**	**1**	**1**	**1**	**1**	**1**	**0**	**1**	**0**	**9**

Columns indicate the result of the *a priori* classification (AC = Alarm Call, CH = Chatter, CL = Click, CO = Group Cohesion Call, H = Hoot, GH = Grunted Hoot, GR = Grunt, GT = Grunt-Tonal Call, LG = Long Grunt, LGA = Alarm Long Grunt, SC = Scream, SN = Snort, TE = Territorial Advertisement Call, TO = Tonal Call), while rows show cluster membership and composition. The number of vocalizations in a cluster (TOT), the number of vocalizations corresponding to an *a priori* vocal type (Input), and the number of exemplars chosen from that vocal type during the cluster analysis (Exemplars) are also reported.

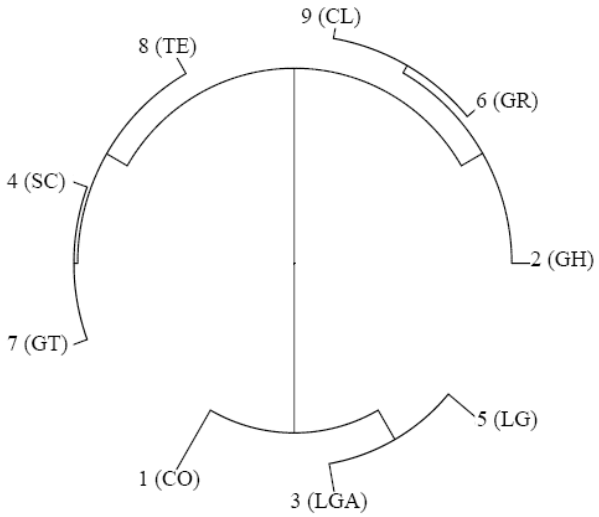


Fig. 12 Schematic representation of the polar dendrogram in Fig. 4 (*E. mongoz*), showing Cluster ID and abbreviation of the vocal type corresponding to the chosen exemplar

Eulemur coronatus

Table SVI. Distribution of the vocalizations of *Eulemur coronatus* in the clusters identified using the affinity propagation algorithm.

Cluster ID	AC	H	GH	GR	GT	SC	TO	TOT
1						5		**5**
2	18						1	**19**
3	2							**2**
4	4					1		**5**
5	2					1		**3**
6				1			11	**12**
7		8	4	20				**32**
8						7		**7**
9		1		13	10		1	**25**
10						11		**11**
11			1	1	12	1	1	**16**
12		7	14	15			2	**38**
13						15		**15**
Input (N)	**26**	**16**	**19**	**50**	**22**	**41**	**16**	**190**
Exemplars (N)	**4**	**0**	**0**	**3**	**1**	**4**	**1**	**13**

Columns indicate the result of the *a priori* classification (AC = Alarm Call, H = Hoot, GH = Grunted Hoot, GR = Grunt, GT = Grunt-Tonal Call, SC = Scream, TO = Tonal Call), while rows show cluster membership and composition. The number of vocalizations in a cluster (TOT), the number of vocalizations corresponding to an *a priori* vocal type (Input), and the number of exemplars chosen from that vocal type during the cluster analysis (Exemplars) are also reported.

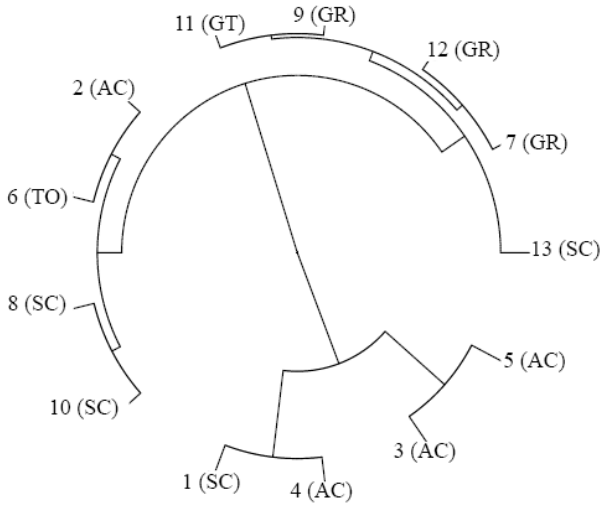


Fig. 13 Schematic representation of the polar dendrogram in Fig. 5 (*E. coronatus*), showing Cluster ID and abbreviation of the vocal type corresponding to the chosen exemplar

Eulemur flavifrons

Table SVII. Distribution of the vocalizations of *Eulemur flavifrons* in the clusters identified using the affinity propagation algorithm.

Cluster ID	AC	CH	H	GH	GR	GT	LG	LGTC	TO	TOT
1	5									**5**
2			15	34	21				5	**75**
3	9									**9**
4		1	5	7			1	1	5	**20**
5		1					1	4	1	**7**
6		5					6	20	2	**33**
7	9									**9**
8	10									**10**
9		1				11	3			**15**
10	6	1								**7**
Input (N)	**39**	**9**	**20**	**41**	**21**	**11**	**11**	**25**	**13**	**190**
Exemplars (N)	**5**	**0**	**0**	**2**	**0**	**1**	**0**	**2**	**0**	**10**

Columns indicate the result of the *a priori* classification (AC = Alarm Call, CH = Chatter, H = Hoot, GH = Grunted Hoot, GR = Grunt, GT = Grunt-Tonal Call, LG = Long Grunt, LGTC = Long Grunt-Tonal Call, TO = Tonal Call), while rows show cluster membership and composition. The number of vocalizations in a cluster (TOT), the number of vocalizations corresponding to an *a priori* vocal type (Input), and the number of exemplars chosen from that vocal type during the cluster analysis (Exemplars) are also reported.

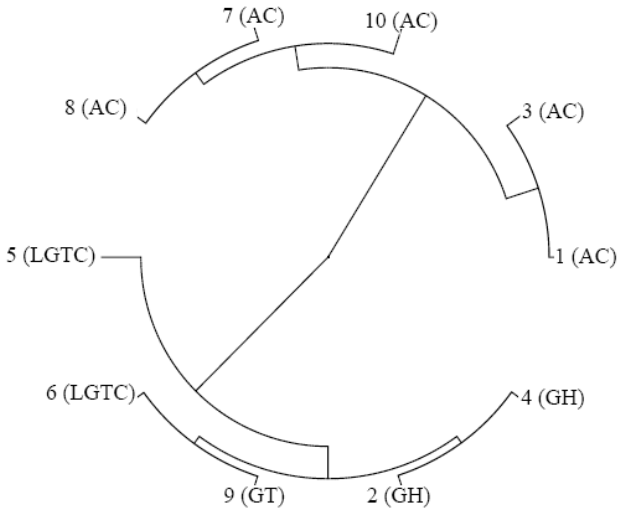


Fig. 14 Schematic representation of the polar dendrogram in Fig. 6 (*E. flavifrons*), showing Cluster ID and abbreviation of the vocal type corresponding to the chosen exemplar

Eulemur macaco

Table SVIII. Distribution of the vocalizations of *Eulemur macaco* in the clusters identified using the affinity propagation algorithm.

Cluster ID	AC	CH	CL	H	GH	GR	LG	LGTC	SC	SN	TO	TOT
1	1								1			**2**
2	1			2			4	1	6	1		**15**
3			12	11	22	39				4	1	**89**
4		1					3	2	7			**13**
5	2	1		1			2		11		3	**20**
6	1	2		1	4	4				2	9	**23**
7							1	3				**4**
8	2											**2**
9	1	2					7	2	2			**14**
10	1								7			**8**
Input (N)	**9**	**6**	**12**	**15**	**26**	**43**	**17**	**8**	**34**	**7**	**13**	**190**
Exemplars (N)	**1**	**0**	**0**	**0**	**0**	**1**	**1**	**1**	**5**	**0**	**1**	**10**

Columns indicate the result of the *a priori* classification (AC = Alarm Call, CH = Chatter, CL = Click, H = Hoot, GH = Grunted Hoot, GR = Grunt, LG = Long Grunt, LGTC = Long Grunt-Tonal Call, SC = Scream, SN = Snort, TO = Tonal Call), while rows show cluster membership and composition. The number of vocalizations in a cluster (TOT), the number of vocalizations corresponding to an *a priori* vocal type (Input), and the number of exemplars chosen from that vocal type during the cluster analysis (Exemplars) are also reported.

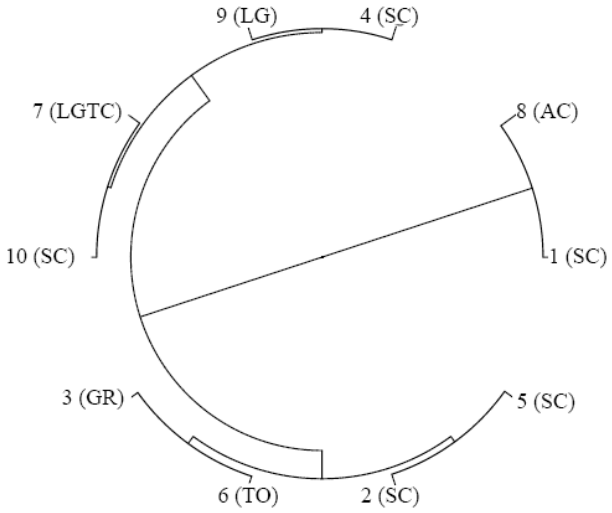


Fig. 15 Schematic representation of the polar dendrogram in Fig. 7 (*E. macaco*), showing Cluster ID and abbreviation of the vocal type corresponding to the chosen exemplar